Object- Oriented Modelling and Programming in Engineering

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Given,

$$P(t) = a1 * t + \sin(2 * \pi * f1 * t)$$

$$a1 = \frac{matriculation\ number(124774)}{40000}\ W$$

$$f = 2Hz$$

For this system, the Calculation of the energy consumption E(t) for t=20 seconds.

$$E(t) = \int_0^t P(t)dt$$

a1 = 3.16985

By subtituting and Solving the Equation

$$P(t) = 3.16985 + \sin(4\pi t),$$

E(t) = 634.469

- Function p(t)

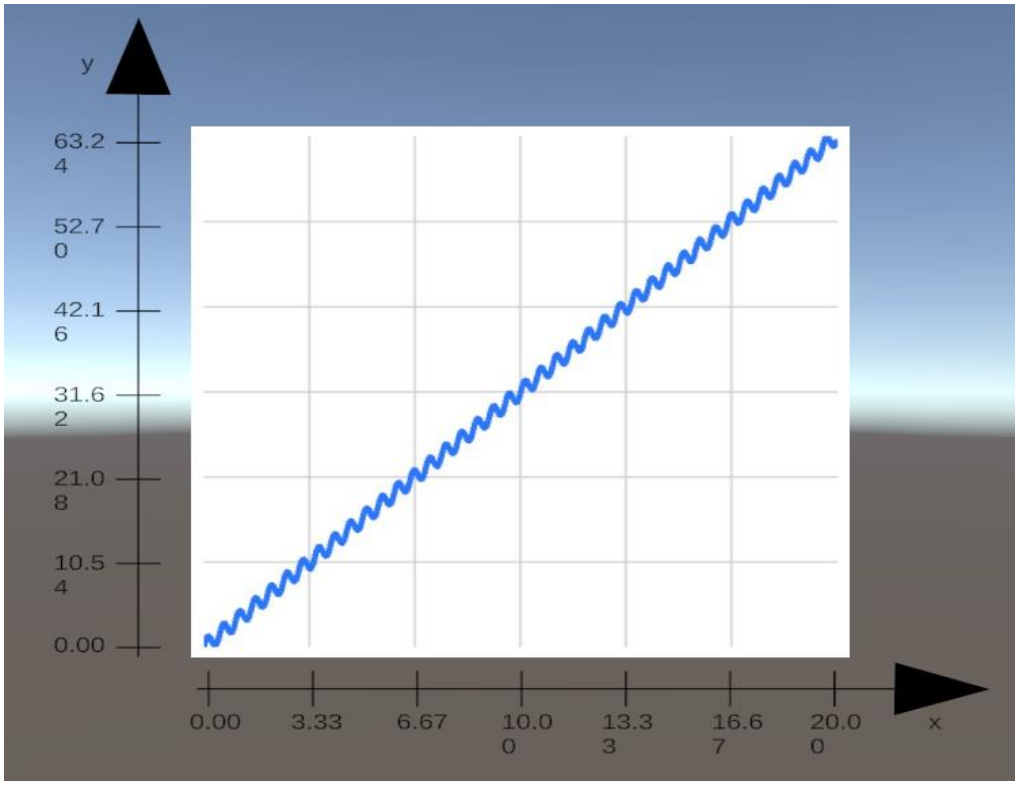


Figure 1: - Function p(t)

- Nassi-Schneiderman diagrams for all three algorithms

Figure 2: Algorithm -1(METHOD –1)

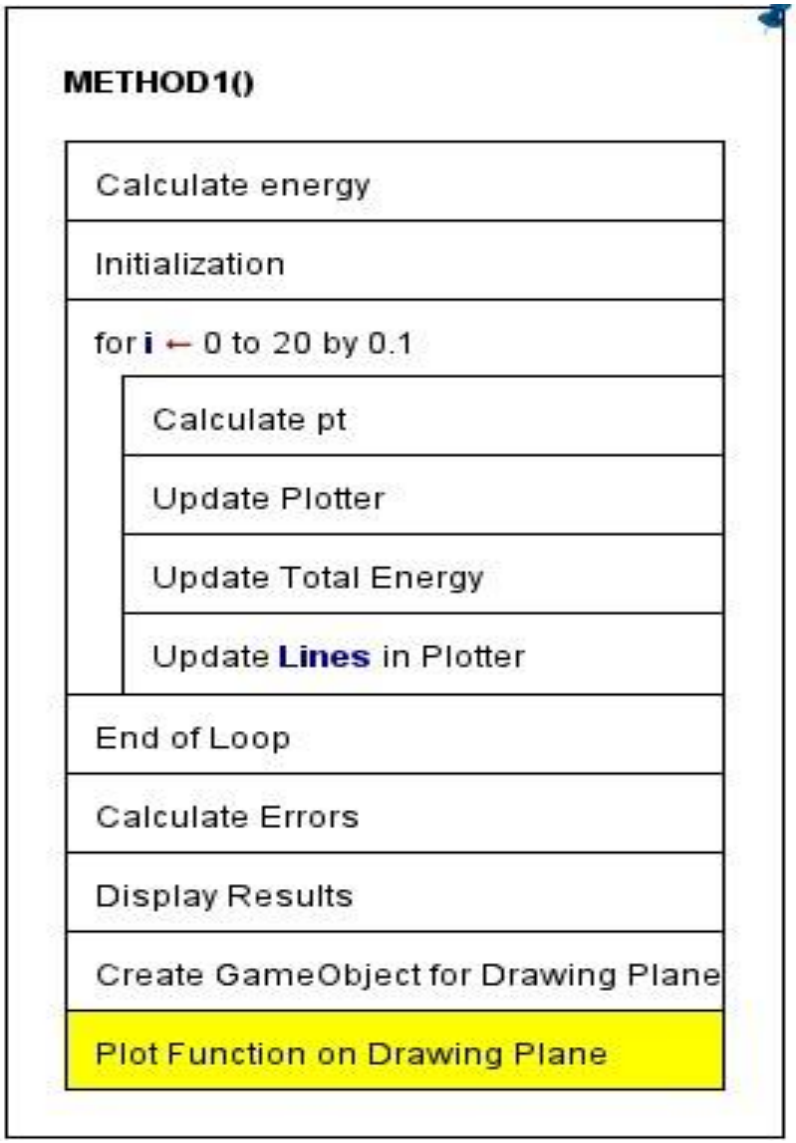


Figure 3: Algorithm -2(METHOD –2)

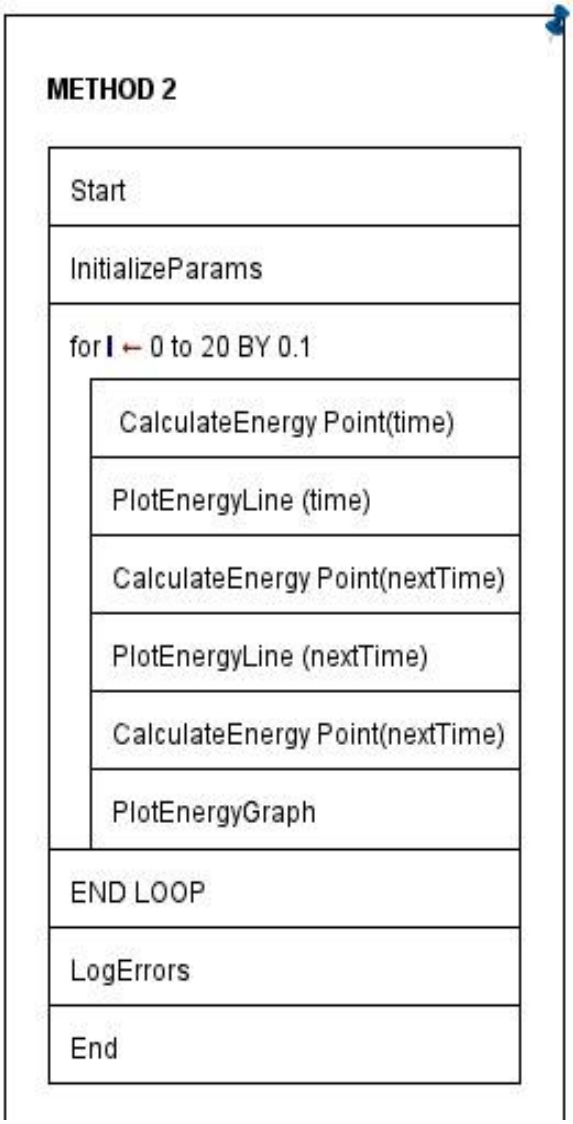
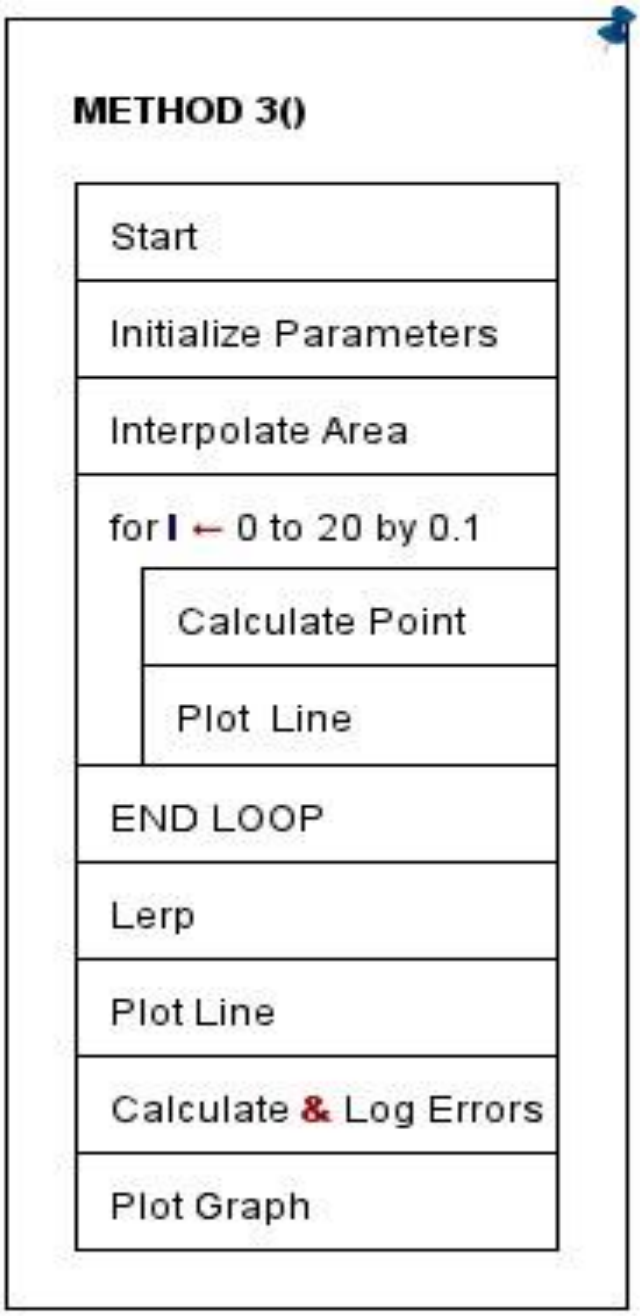
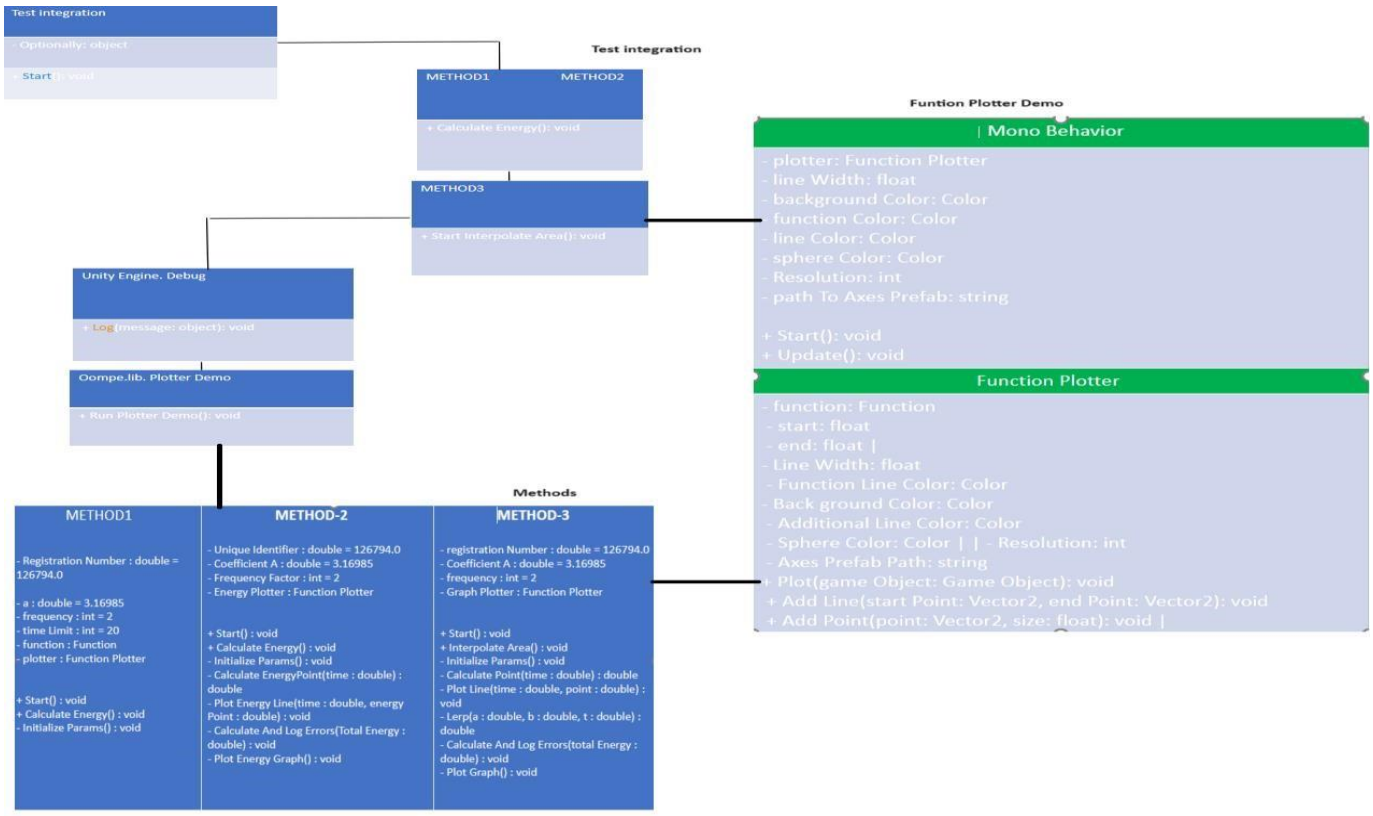


Figure 4: Algorithm -3(METHOD –3)



- UML- Diagram

Figuer-5 UML-Diagram



- Results for algorithm

  Plot for Algorithm-1(Method-1)

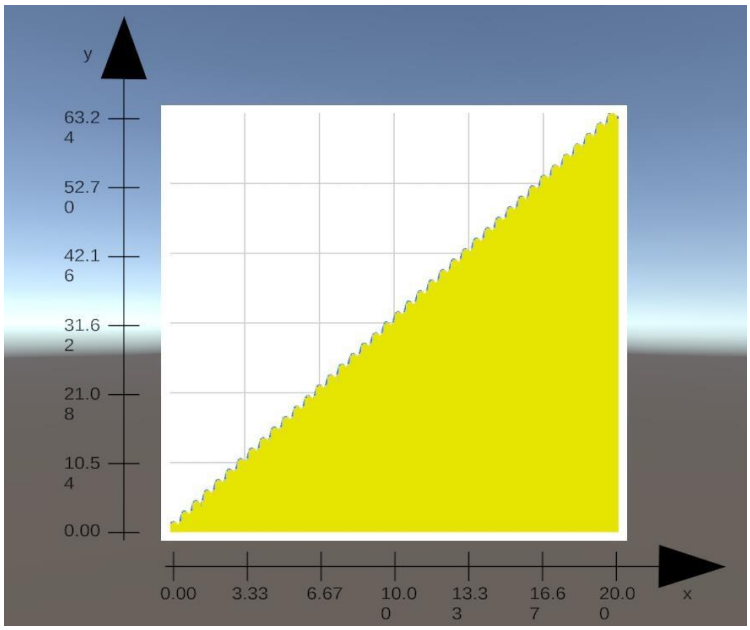


Figure-6 Plot for Algorithm-1(Method-1) For 20 seconds with grid size of 0.1

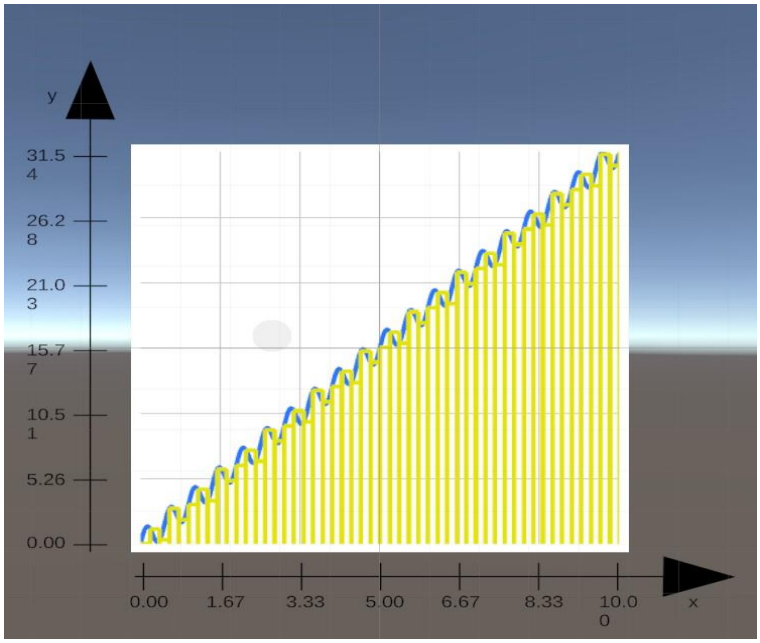


Figure-7 Plot for Algorithm-1(Method-1) For 10 seconds with grid size of 0.2

Plot for Algorithm-2(Method-2)

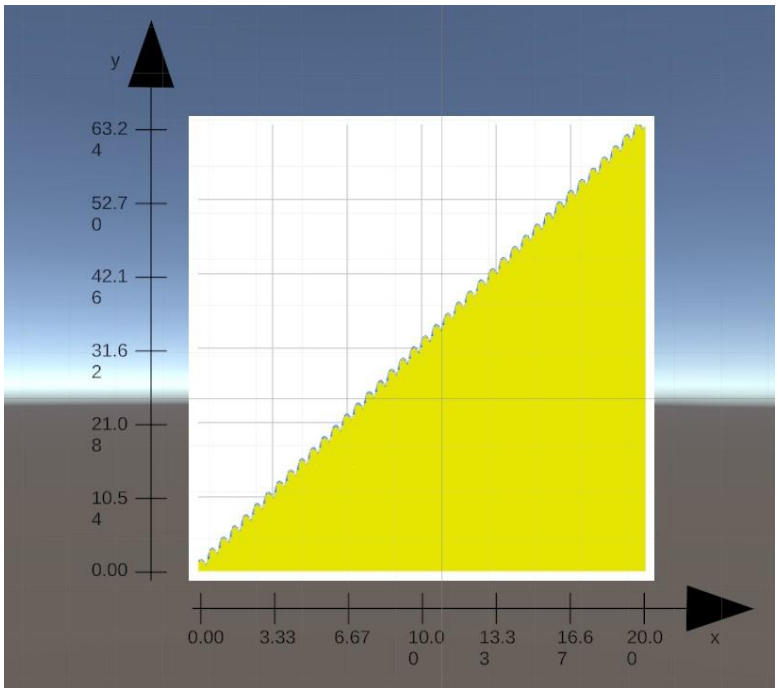


Figure-8 Plot for Algorithm-2(Method-2) For 20 seconds with grid size of 0.1

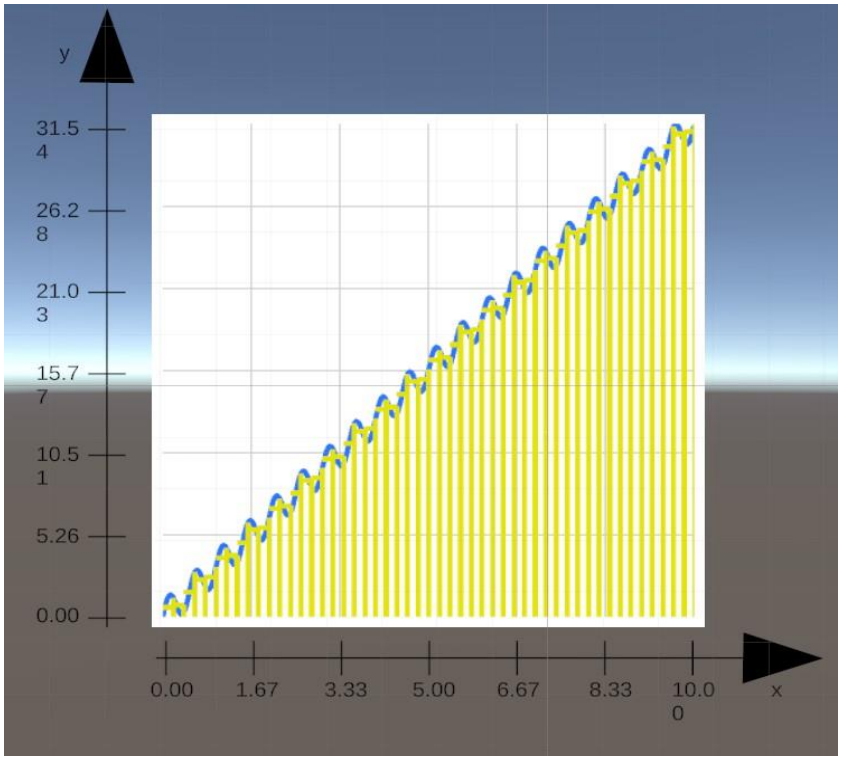


Figure-9 Plot for Algorithm-2(Method-2) For 10 seconds with grid size of 0.2 Plot for Algorithm-3(Method-3)

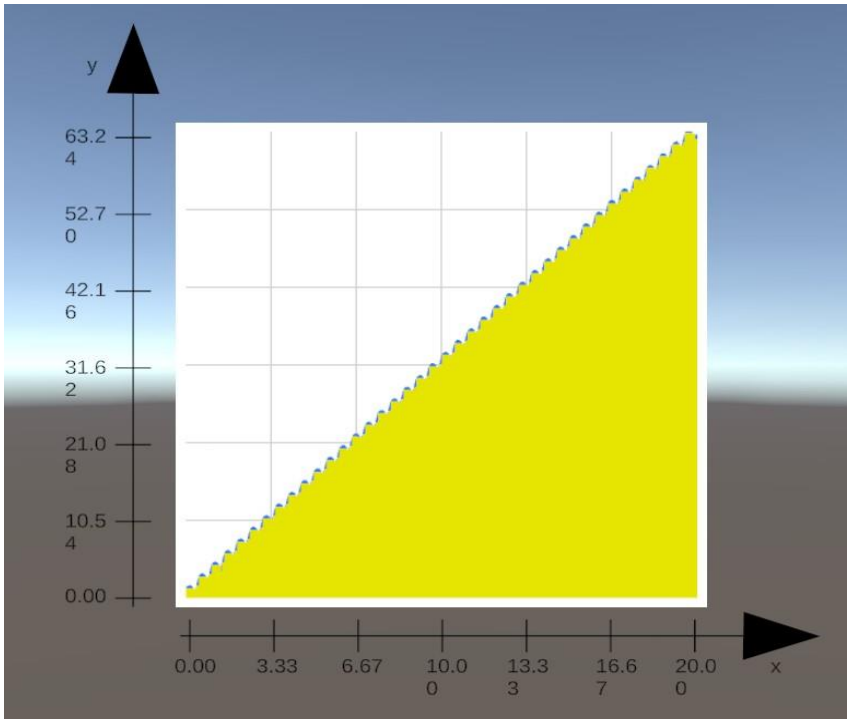


Figure-10 Plot for Algorithm-3(Method-3) For 20 seconds with grid size of 0.1

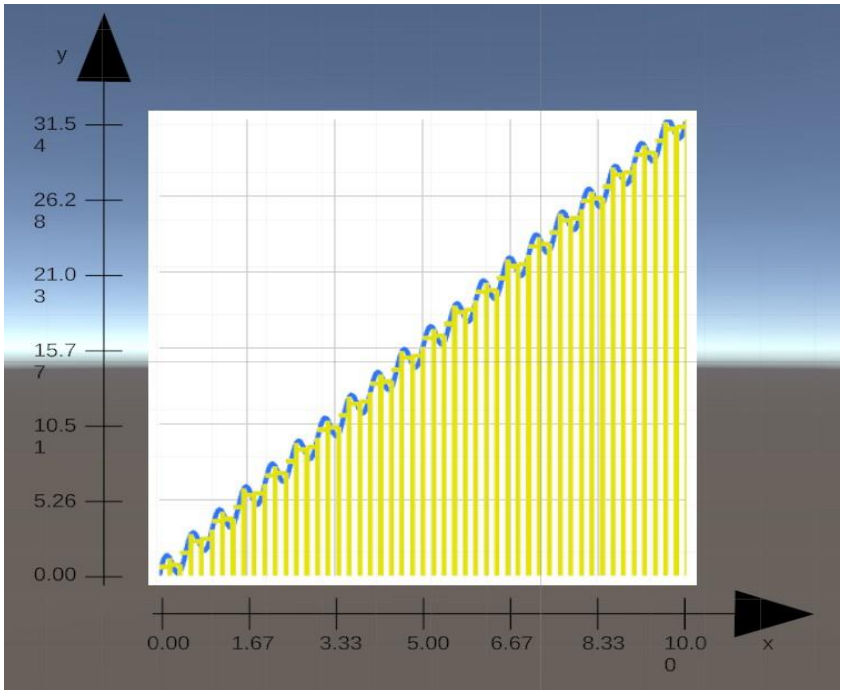


Figure-11 Plot for Algorithm-3(Method-3) For 10 seconds with grid size of 0.2

- Absolute and Relative Error

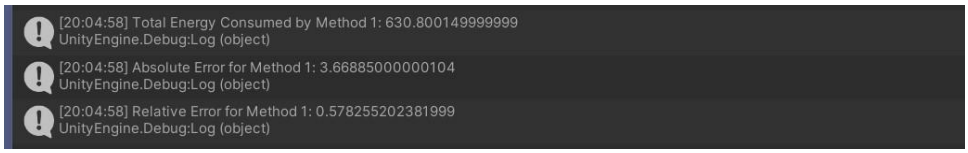


Figure-12 Algorithm-1(Method-1) For 20 seconds with grid size of 0.1



Figure-13 Algorithm-1(Method-1) For 10 seconds with grid size of 0.2

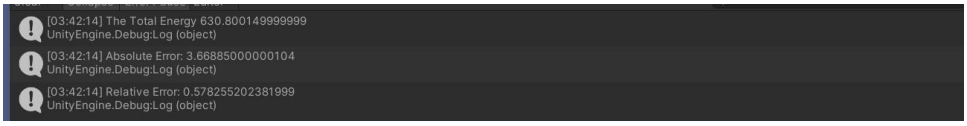


Figure-14 Algorithm-2(Method-2) For 20 seconds with grid size of 0.1

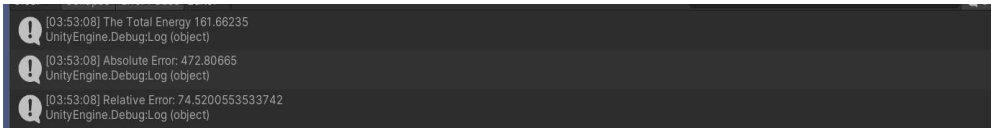


Figure-15 Algorithm-2(Method-2) For 10 seconds with grid size of 0.2

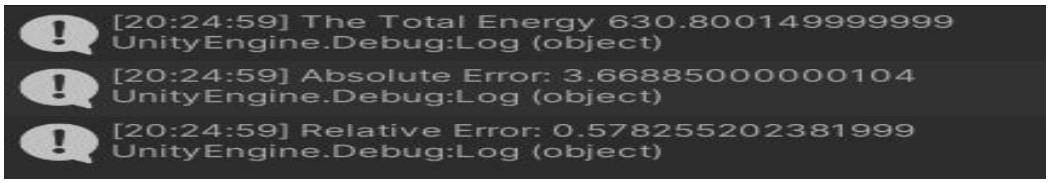


Figure-16 Algorithm-3(Method-3) For 20 seconds with grid size of 0.1

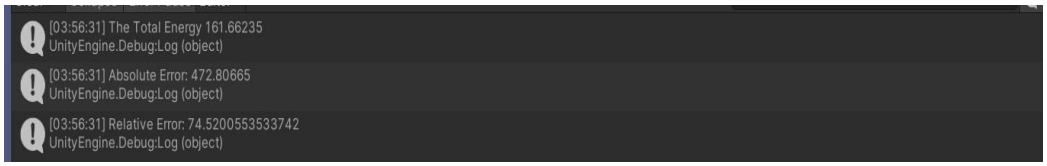


Figure-17 Algorithm-3(Method-3) For 10 seconds with grid size of 0.2

References

1) Lecture slides and Lab class codes